

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept understanding Analyser
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP8-style 4-space indentation used throughout all .py files
2	Proper File Structure	Files and folders are logically organized	Yes	Flat modular structure: app.py, speech_to_text.py, audio_utils.py, semantic_eval.py, scoring_engine.py, report_generator.py
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. transcript, similarity, filler_ratio, audio_feats, final_score
4	Function / Method Names	Functions are descriptively named	Yes	e.g. speech_to_text(), extract_audio_features(), evaluate_understanding(), generate_pdf_report()
5	Code Comments	Inline and block comments explain logic	Yes	Each module has a docstring explaining its purpose; inline comments on scoring logic
6	Modular Design	Code is split into reusable functions/modules	Yes	Each responsibility separated into its own file (transcription, audio analysis, semantic evaluation, scoring, reporting, UI)
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate functions; shared logic centralized per module
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic checks present (e.g. empty transcript); could be expanded for corrupted/unsupported audio files

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	speech_to_text()	Python (Whisper)	Called for every audio upload	High
2	extract_audio_features()	Python (Librosa)	Reusable for any audio file	High
3	compute_similarity()	Python (Sentence-BERT)	Reusable for any transcript/reference pair	High
4	evaluate_understanding()	Python	Pure function, reusable independent of UI	High
5	generate_pdf_report()	Python (ReportLab)	Reusable for any report content	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean modular file separation, no monolithic files
Readability	4	Clear naming; could add more inline comments
Reusability	4	Core functions are stateless and reusable outside this app
Documentation / Comments	3	Module-level docstrings present; could add more detail
Overall Score	4	Solid, maintainable codebase overall